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50/100

CHEMICAL REACTION ENGINEERING

FINAL EXAM

CHE 321, SPRING 2025, CLOSED BOOK, TIME: 60 MIN

The endothermic gas phase reaction $A \rightarrow B + C$ is carried out in a single steam-jacketed fluidized bed reactor, where the temperature of the steam circulating through the jacket remains constant at $T_a = 300°C$.

For the following data, calculate the steady-state reactor temperature.

Conversion: $X = 85\%$
Heat-transfer area of the jacket: $A = 5\ m^2$
Overall heat transfer coefficient of the jacket: $U = 500 \frac{J}{s.m^2.K}$
Heat of reaction per unit mole of species A at 25°C: $\Delta_r H = 120 \frac{kJ}{mol}$
Feed temperature: $T_0 = 40\ °C$
Feed molar flow rates: $F_{A0} = 10 \frac{mol}{s}$, $F_{I0} = 2 \frac{mol}{s}$ (inert), $F_{B0} = F_{C0} = 0$
Heat capacities (assume constant) $c_{pA} = 90 \frac{J}{mol.K}$, $c_{pB} = 60 \frac{J}{mol.K}$, $c_{pC} = 40 \frac{J}{mol.K}$, $c_{pI} = 50 \frac{J}{mol.K}$
Reactor energy balance: $\frac{dU}{dt} = \sum_\alpha F_{\alpha 0}\, h_\alpha(T_0) - \sum_\alpha F_\alpha\, h_\alpha(T) + \dot{Q} + \dot{W}_s$

Conversion $X \equiv \frac{F_{A0} - F_A}{F_{A0}}$

PBR $\quad T_a = 300°C$

PBR mole balance

$$F_{A0} - F_A + \int^V r_A dV = 0 \Rightarrow \frac{dF_A}{dV} = -r_A$$

$$F_{A0}\frac{dX}{dW} = -r_A \qquad \frac{dX}{dV} = \frac{-r_A}{F_{A0}}$$

Energy balance $\frac{dU}{dt} = \sum_\alpha F_{\alpha 0} h_\alpha(T_0) - \sum_\alpha F_\alpha h_\alpha(T) + Q + \dot{W}_s$

PBr so no work $\dot{W}_s = 0$

$$\dot{Q} - \dot{W}_s - F_{A0}\sum \Theta_i C_{pi}(T - T_0) - F_{A0}X = 0$$

$$0 = \sum_\alpha F_{\alpha 0} h_\alpha(T_0) - \sum_\alpha F_\alpha h_\alpha(T) + \dot{Q} + \dot{W}_s$$

$$\Rightarrow \dot{Q} - F_{A0}\sum \Theta_i C_{pi}(T - T_0) - F_{A0}X = 0$$

$$Q_a = \dot{m}_a c_{pa}(T_a - T_{a0})$$

$\dot{Q} - F_{AO} \sum \Theta_i C_{pi} (T - T_O) - F_{AO} X = 0$

$\dot{Q} = U * A * \Delta T$

$\dot{Q} = m c_p * (T_{aO} - T_a)$

$U = 500 \frac{J}{s m^2 K} \cdot 5 m^2 * \Delta T$

$\dot{Q} = 500 \frac{J}{s m^2 K} \cdot 5 m^2 (T - T_O)$

$UA\Delta T = UA(T - T_O)$

$UA(T - T_O) - F_{AO} \sum \Theta_i C_{pi} (T - T_O) - F_{AO} X = 0$

$0 = \sum F_{\alpha o} h_\alpha (T_O) - \sum F_\alpha h_\alpha (T) + \dot{Q} = 0$

$\Rightarrow 0 = \sum F_{\alpha o} h_\alpha (T_O) - \sum F_\alpha h_\alpha (T) + (UA (T_{ao} - T_a)$ ✓

Find FA values from $X = \frac{F_{AO} - F_A}{F_{AO}}$ $X = F_{AO} - F$

$\sum F_{\alpha o} h_\alpha (T_O)$

$\Rightarrow = 10 \frac{mol}{s}$

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$\Delta_r H_A = 120 \frac{kJ}{mol}$

calculating h_α

$h_\alpha = Cp_\alpha \cdot T_0$

$h_\alpha (\frac{J}{mol}) = \frac{J}{mol\,K} \cdot K$

$T_0 = 40°C$. $T = 313K$

$h_A = 90 \frac{J}{mol K} \cdot 313\,K = 28170 \frac{J}{mol}$

$h_B = 60 \frac{J}{mol K} \cdot 313 K = 18780 \frac{J}{mol}$

$h_C = 40 \frac{J}{mol K} \cdot 313 K = 12520 \frac{J}{mol}$

$h_I = 50 \frac{J}{mol K} \cdot 313 K = 15650 \frac{J}{mol}$

$F_T = F_A + F_B + F_C$

$-r_A =$

$C_A = C_{T0} \frac{F_A}{F_T} P \frac{T_0}{T}$

$C_A = \frac{kmol}{dm^3} \cdot \frac{dm^3}{s}$

$C_A v_0 = F_A$

$X = \frac{F_{A0} - F_A}{F_{A0}}$

$.85 = \frac{(10 \frac{mol}{s}) - F_A}{(10 \frac{mol}{s})} \Rightarrow F_A = 1.5 \frac{kmol}{s}$

We can write F_B and F_C in terms of conversion

$0.85 = \frac{F_{B0} - F_B}{F_{B0}}$

$\frac{dF_B}{dV} = r_A$

$\frac{dX}{dV} = \frac{-r_B}{F_{B0}}$

$$\frac{dx}{dV} = \frac{-r_A}{F_{A0}} \qquad 0.85 = \frac{-r_A}{F_{A0}} dV$$

$$0.85 =$$

relate T and conversion

$$T = T_0 + \frac{(-\Delta H_{rx})X}{\sum \Theta_i C_{pi}} + \dot{Q}$$

$$\Theta = \sum_{\Theta} F$$

$$T = T_0 + \frac{(-\Delta H_{rx})X}{\sum \Theta_i C_{pi}} + UA(T - T_0)$$

$$T_0 = 40^\circ C \qquad \Theta_i = \frac{F_{i0}}{F_i}$$

$$\Theta_A = 1 \quad \Theta_B = 0 \quad \Theta_C = 0$$

$$\Theta_I = \frac{2}{5}$$

$$T = 313K + \frac{\left(120000 \frac{J}{mol}\right) 0.85}{(1)\left(90 \frac{J}{mol K}\right) + \left(\frac{2}{5}\right)\left(50 \frac{J}{mol K}\right)} \left(500 \frac{J}{s m^2 K}\left(T - 313K\right)\right)$$

$$\boxed{T = 313K}$$

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